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**BACHELOR OF INFORMATION SYSTEM (HONS.) INFORMATION SYSTEM
ENGINEERING**

**CSC584 ENTERPRISE PROGRAMMING GROUP PROJECT
TITLE: FOOD BANK & SOCIAL VOUCHER MANAGEMENT SYSTEM
PREPARED BY**

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1.0 Introduction

The Food Bank & Social Voucher System is a web-based application developed to assist the management of food assistance services within a campus environment. The system is designed to support HEP (Hal Ehwal Pelajar) staff in managing food donations, monitoring inventory, and distributing food items and social vouchers to students who are in need of assistance.

In many institutions, food bank management is still carried out manually using spreadsheets or paper-based records. This traditional approach often leads to several issues such as data inconsistency, difficulty in tracking donation records, and inefficiency in generating reports. In addition, manual systems make it challenging for administrators to obtain real-time information regarding food stock levels and distribution activities.

Therefore, this system is developed to digitalize and centralize all food bank operations into a single platform. By using a web-based system, all data related to donations, inventory, and distribution can be stored in a structured database and accessed in real time. This improves the efficiency, accuracy, and transparency of the food assistance process within the campus.

The system is developed using Java Web Technology and follows the Model-View-Controller (MVC) architecture, which separates the system into three main components: JavaBeans as the Model, JSP as the View, and Servlet as the Controller. This architecture ensures that the system is well-structured, scalable, and easy to maintain.

The system is expected to improve efficiency, reduce manual errors and also to provide better transparency in managing campus food assistance services.

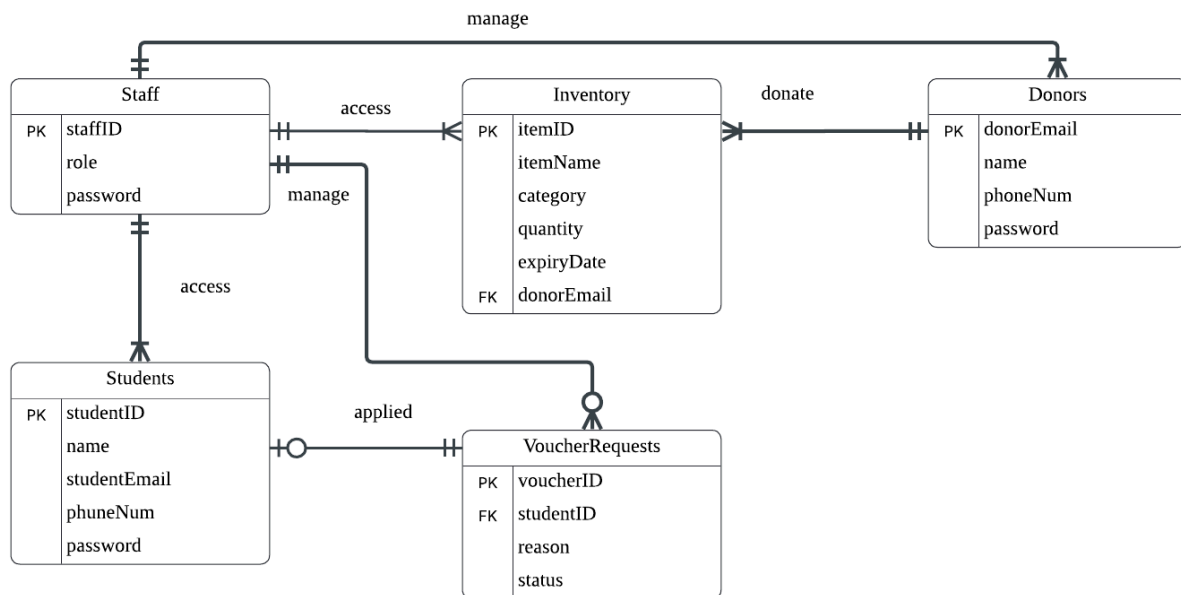
2.0 Problem Statement

The current manual food bank management system faces major challenges because it relies on paper-based and disorganized processes. Manual record-keeping raises the risk of losing data and creating inconsistencies. Without a centralized system, HEP staff cannot monitor food inventory levels in real time. Additionally, the lack of a clear tracking method for food and voucher distribution makes it hard to accurately monitor student beneficiaries. These manual issues lead to inefficient reporting and analysis due to the absence of immediate data access. This ultimately wastes valuable time and effort and reduces the staff's overall efficiency in managing food assistance programs.

3.0 Objective

- To design and implement the system using the MVC (Model-View-Controller) pattern within a Java environment, ensuring a structured, scalable, and maintainable codebase.
- To design a robust database schema using entity relationship diagram (ERD) and integrate Apache Derby as the relational database management system for secure, persistent data storage.
- To implement secure access control that manages distinct permissions for Admins, Staff, Students, and Donors, ensuring data privacy and restricted system access.
- To digitize and automate manual workflows, specifically regarding inventory tracking and the processing of food voucher requests.
- To maintain long-term system accuracy through automated relational constraints, such as cascading deletes for user profiles and data anonymization for donated inventory.
- To provide an accessible web interface that enables stakeholders to easily manage their profiles, track contributions, and monitor system data in real-time.

4.0 Entity Relationship Diagram (ERD)



5.0 User Manual

i. Login Page

- **Enter your Username / Email Address:** Input your username or email address into the username field.
- **Enter your Password:** Input your password
- **Click Login:** Once the information is entered correctly, click the “Login” button.
- **Password Recovery:** If you forget your password, contact the number or email in the Need Help section.

Welcome - UniCare Food Portal

localhost:8080/CSC584-UniCareFood-Portal-main/index.jsp

UniCare Food Portal
Care Shared, Futures Fed

Username / Email Address
admin

Password

Authenticate & Login

Need an account? [Register Here](#)

Need Help?
Forgot your password or experiencing login issues?
Please contact the UniCare Food Portal Administrator.
+60 3-5544 1234
admin@unicarefood.my

Our team will verify your identity before assisting with password recovery.

ii. Signup Page

If Donor:

- **Choose your Account Type:** Select either Donor or Student in the Account Type field.
- **Fill in your Full Name and Email:** Input your full name and a valid email address.
- **Fill in your Phone Number:** Input your own and valid phone number.
- **Create a Password:** Choose a strong password for security.

Public Registration - UniCare Food Portal

localhost:8080/CSC584- UniCareFood-Portal-main/public_register.jsp

Create an Account

Join the UniCare Food Portal community

Account Type

Donor

Full Name

e.g. Amirul Bakhtiar

Email Address (Will be your Login ID)

name@example.com

Phone Number

e.g. 0123456789

Password

Minimum 5 characters

[Register](#)

[Back to Login](#)

If Student:

- **Choose your Account Type:** Select either Donor or Student in the Account Type field.
- **Fill in your Full Name and Email:** Input your full name and a valid email address.
- **Fill in your Student ID:** Input your own student identification number.
- **Fill in your Phone Number:** Input your own and valid phone number.
- **Create a Password:** Choose a strong password for security.

Public Registration - UniCare Food Portal

localhost:8080/CSC584- UniCareFood-Portal-main/public_register.jsp

Create an Account

Join the UniCare Food Portal community

Account Type

Student

Full Name

e.g. Amirul Bakhtiar

Student ID (Will be your Login ID)

e.g. S12345

Email Address

name@example.com

Phone Number

e.g. 0123456789

Password

Minimum 5 characters

[Register](#)

Admin & Staff Interface

iii. Dashboard

Admin Tools

- **Manage All Users:** Click the Manage All Users button to view, edit, or remove registered accounts across the platform.
- **Register New Staff:** Click the Register New Staff button to onboard and create credentials for new administrative or inventory team members.

Quick Navigation Management Buttons

- **Manage Inventory:** Instantly routes you to the main inventory controls to track, add, or edit food supplies and product categories.
- **Manage Voucher Requests:** Takes you straight to the incoming application queue where you can review, approve, or deny student requests.
- **Manage Students:** Opens the student tracking database to verify registrations and control voucher allocations.
- **My Profile:** Click this to update your staff account details, contact info, and security credentials.

Food Bank Dashboard Overview

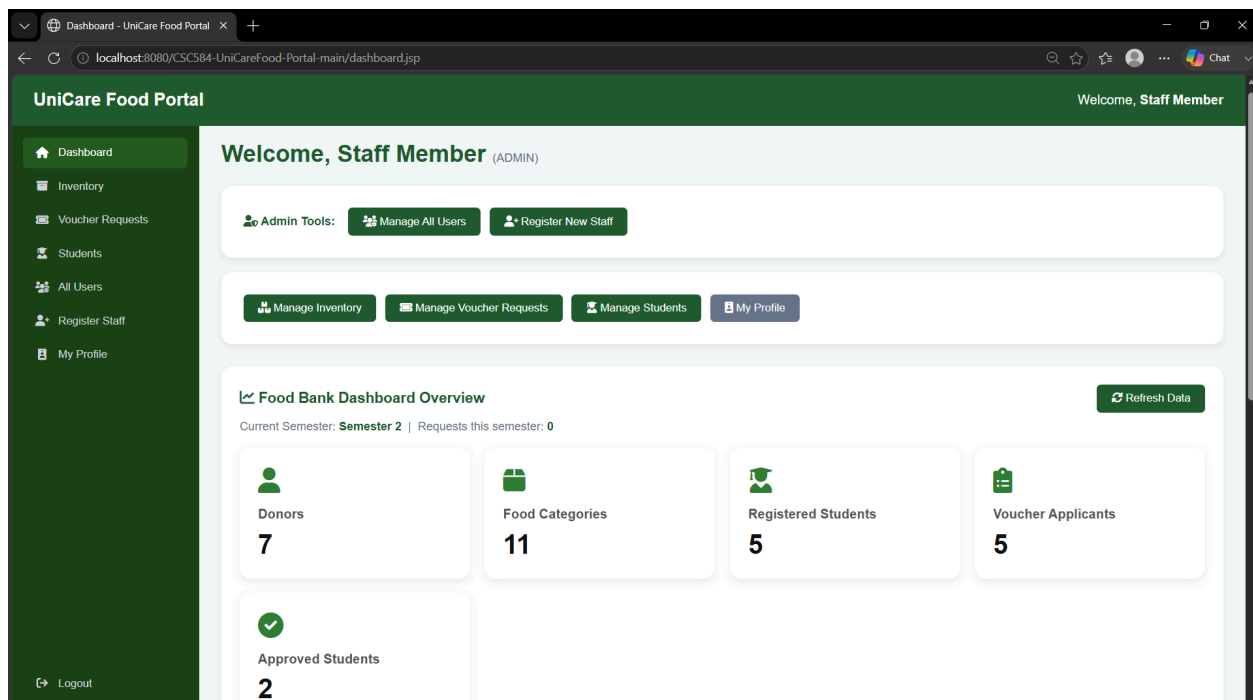
- **View Real Time Metrics:** Monitor the total count of Donors, Food Categories, Registered Students, and Voucher Applicants, and Approved Students.
- **Check Semester Stats:** Track the current academic term alongside total processing requests logged during this period.
- **Refresh Data:** Click the Refresh Data button to reload the dashboard metrics with live updates from the database.

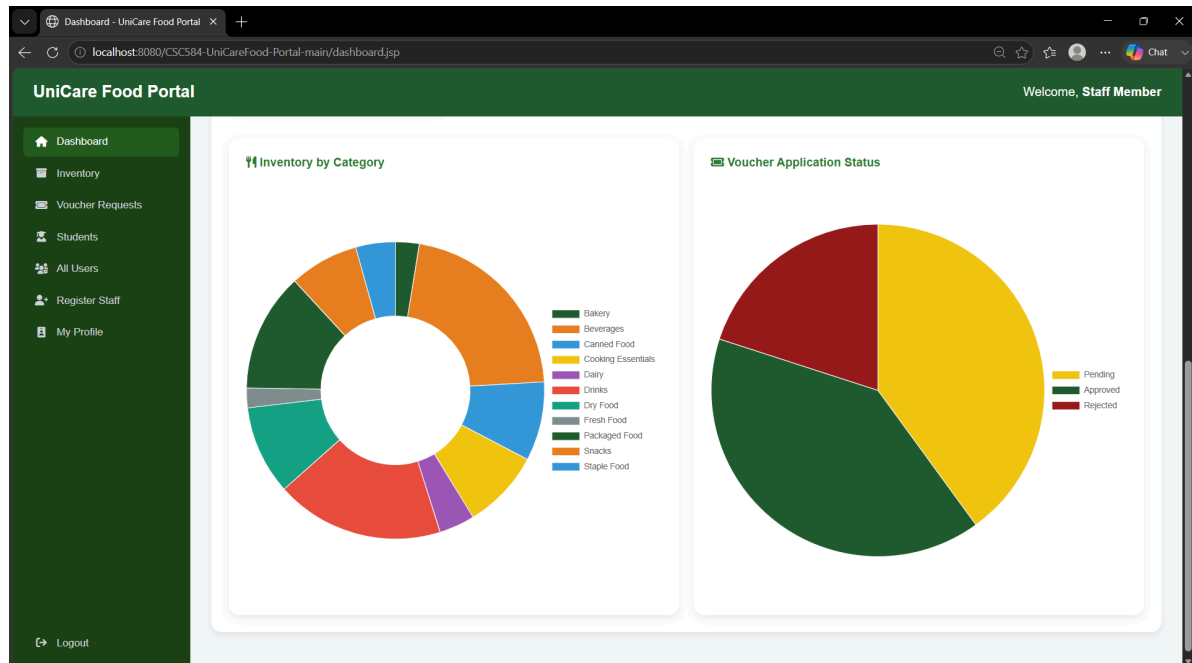
Dashboard Analytics & Charts

- **Inventory by Category:** This donut chart dynamically visualizes stock distribution across various food groups. Use the color-coded legend to check breakdown levels for

items such as Bakery, Beverages, Canned Food, Cooking Essentials, Dairy, Drinks, Dry Food, Fresh Food, Packaged Food, Snacks, and Staple Food.

- **Voucher Application Status:** This pie chart gives you an instant diagnostic breakdown of all submitted student meal voucher applications. It tracks and highlights the total of student requests currently sitting across three primary administrative states:
 - a. **Pending:** Applications awaiting staff review.
 - b. **Approved:** Completed requests successfully processed.
 - c. **Rejected:** Denied submissions.





iv. Inventory Page

Inventory Metrics Cards

- **Total Items (Qty):** You can view the total aggregated quantity of all food items stored in the system.
- **Food Types:** You can see the total number of distinct food categories registered in the portal.
- **Low Stock (<10):** You can track items that have dropped below a critical threshold of 10 units for immediate attention.
- **Near Expiry (≤ 7 days):** You can identify items that are expiring within a week so you can distribute them urgently.

Filter & Search Controls

- **Category Search:** You can enter a specific type of food to narrow down stock lists by group.
- **Expiry Date Range:** You can set an *Expiry From* and *Expiry To* calendar window to isolate items with specific shelf lives.

- **Item Name:** You can search for specific food items directly by typing their name into this field.
- **Donor (name/email):** You can track down specific inventory contributions by searching for a donor's name or contact email.
- **Apply Filters & Clear Buttons:** You can click Apply Filters to update the results in the table, or click Clear to reset your search criteria.

Current Inventory Data Table

- **View Stock Records:** You can browse real-time details including item names, categories, quantities, expiry dates, and donor contact information.
- **Edit Item:** You can click the green Edit button to modify an item details or adjust its current quantity.
- **Delete Item:** You can click the red Delete button to permanently remove an item record from the active inventory.

UniCare Food Portal Welcome, Staff Member

Food Bank Inventory Management

- Total Items (Qty): **465**
- Food Types: **11**
- Low Stock (<10): **0**
- Near Expiry (<=7 days): **0**

Category: Expiry From: Expiry To:

Item Name: Donor (name/email): Apply Filters Clear

Current Inventory

Item	Category	Qty	Expiry Date	Donor Name	Donor Email	Donor Phone	Actions
Eggs (30 Pack)	Fresh Food	10	2026-07-25	Mohd Hafiz	hafiz@gmail.com	0197654321	Edit Delete
Sugar (1kg)	Cooking Essentials	25	2028-07-31	Amy Lim	amy.lim@gmail.com	0171234567	Edit Delete
Milk Powder	Dairy	18	2028-02-10	John Tan	john.tan@email.com	0169088276	Edit Delete

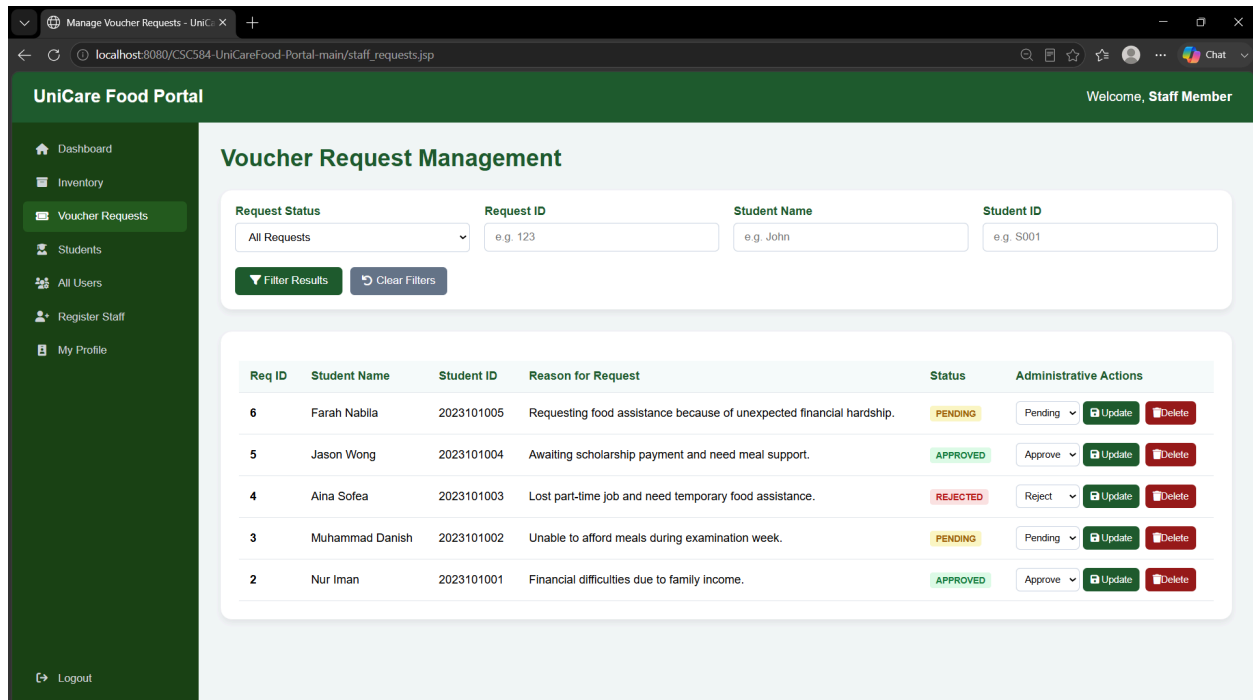
v. Staff Request Page

Search & Filter Panel

- **Request Status Dropdown:** You can select a status (such as *All Requests*, *Pending*, *Approved*, or *Rejected*) to narrow down the application list.
- **Request ID:** You can type a specific number to quickly pull up an individual request record.
- **Student Name:** You can search for applicants by typing a student's name into this field.
- **Student ID:** You can look up applications using a specific university student identification number.
- **Filter & Clear Buttons:** You can click the green Filter Results button to display your customized search, or click Clear Filters to reset all fields back to default.

Request Management Table

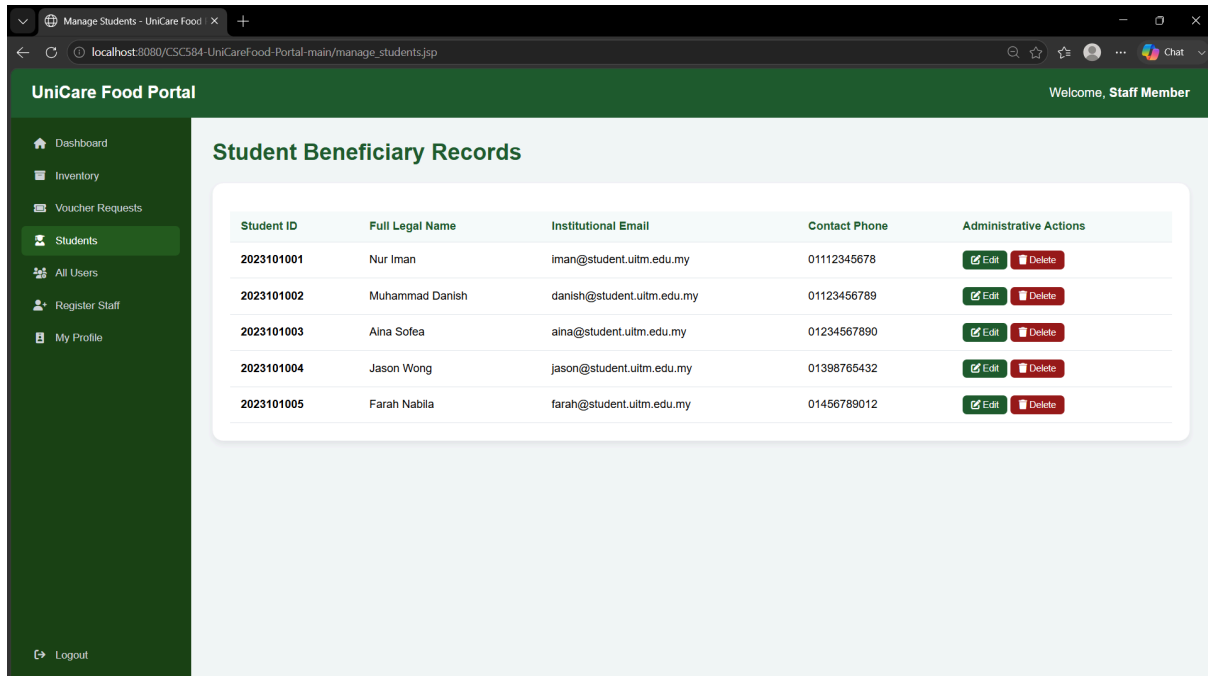
- **Review Student Submissions:** You can view essential applicant details including the request ID, student name, student ID, and the student's stated reason for requesting assistance.
- **Check Status Labels:** You can instantly see the current state of each submission labeled as PENDING, APPROVED, or REJECTED.
- **Update Request Status:** You can select a new status from the dropdown menu and click the green Update button to save changes and process the students voucher.
- **Delete Record:** You can click the red Delete button to permanently remove a voucher request record from the portal database.



vi. Manage Student Page (Admin)

Student Records Table

- **View Student Profiles:** You can see essential student details including their Student ID, Full Name, Email, Contact Phone number, and Reason for Request.
- **Edit Student Details:** You can click the green Edit button to modify an individual student's profile information or account parameters.
- **Delete Student Record:** You can click the red Delete button to permanently remove a student's profile and registration data from the portal.



vii. Manage User Page (Admin)

Search & Filter Panel

- **Functional Role View:** You can filter the directories by choosing specific user groups or choosing to show all tables at once.
- **Keyword Database Query:** You can search for specific system users by typing an identifier, name string, or email address into this field.
- **Apply Filters & Clear Buttons:** You can click Apply Filters to refresh the displayed tables or click Clear to remove your active search conditions.

Staff & Administrators Directory

- **View Staff Credentials:** You can audit operational profiles to check their logged identity tags and security authorization badges (such as ADMIN or STAFF).
- **Manage Account Modifications:** You can click the green Edit button to adjust permissions or click the red Delete button to remove profile credentials.

- **Current Session Protection:** You can see which profile is actively in use marked as your Current Session Account, which safely disables self-deletion to prevent accidental lockouts.

Philanthropic Donors Directory

- **View Contributor Details:** You can scan active lists of external portal donors including their primary email keys, full corporate or personal names, and mobile contact lines.
- **Modify Donor Profiles:** You can click the green Edit button to fix donor registration data or click the red Delete button to drop them from the system directories.

Registered Student Beneficiaries Directory

- **View Beneficiary Profiles:** You can review all enrolled student recipient accounts detailing their specific matriculation identity tags, full legal names, institutional emails, and phone contacts.
- **Update Student Entries:** You can click the green Edit button to reconfigure structural student accounts or click the red Delete button to cancel their profile registration.

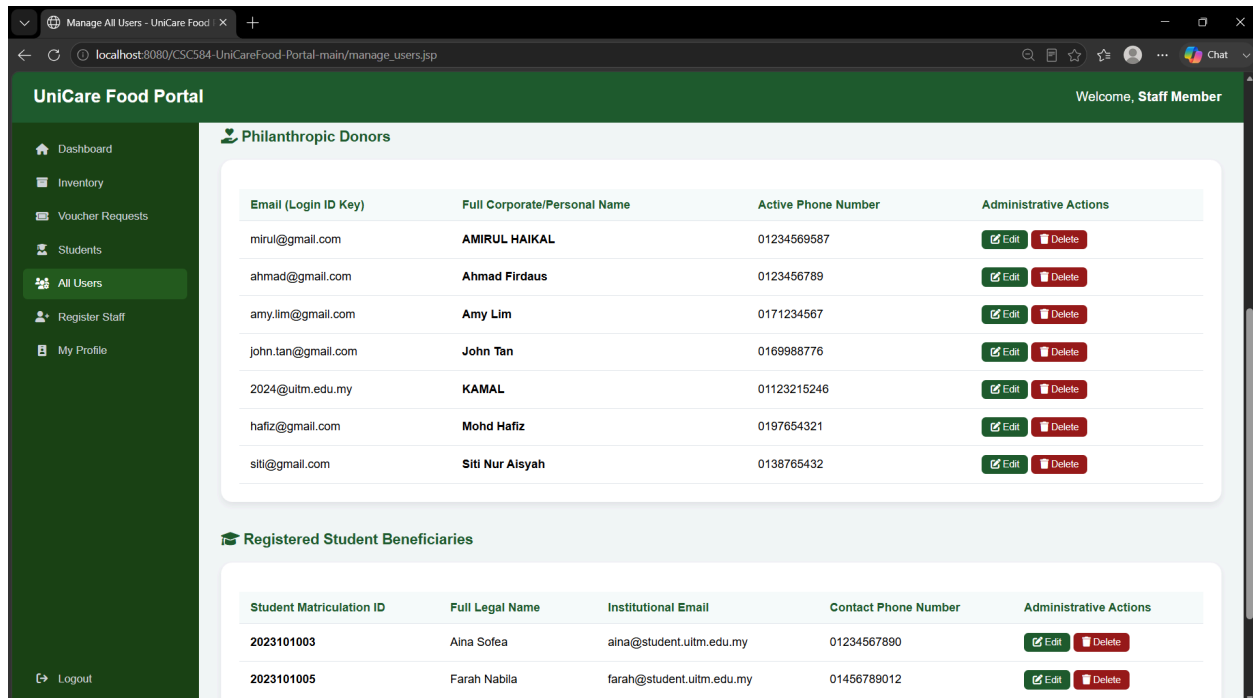
The screenshot shows the 'UniCare Food Portal' interface. The left sidebar contains navigation links: Dashboard, Inventory, Voucher Requests, Students, All Users (highlighted), Register Staff, and My Profile. The main content area is titled 'User Account Directory Management' and includes a 'Functional Role View' dropdown set to 'Display All Sub-Tables' and a 'Keyword Database Query' search bar. Below this, there are two sections: 'Staff & Administrators' and 'Philanthropic Donors'.

Staff & Administrators Table:

Staff ID Identifier	Assigned Security Role	Administrative Actions
ADM001	ADMIN	[Edit] [Delete]
admin	ADMIN	[Edit] [Current Session Account]
STF001	STAFF	[Edit] [Delete]
STF002	STAFF	[Edit] [Delete]

Philanthropic Donors Table:

Email (Login ID Key)	Full Corporate/Personal Name	Active Phone Number	Administrative Actions
mirul@gmail.com	AMIRUL HAIKAL	01234569587	[Edit] [Delete]



viii. Staff Register Page

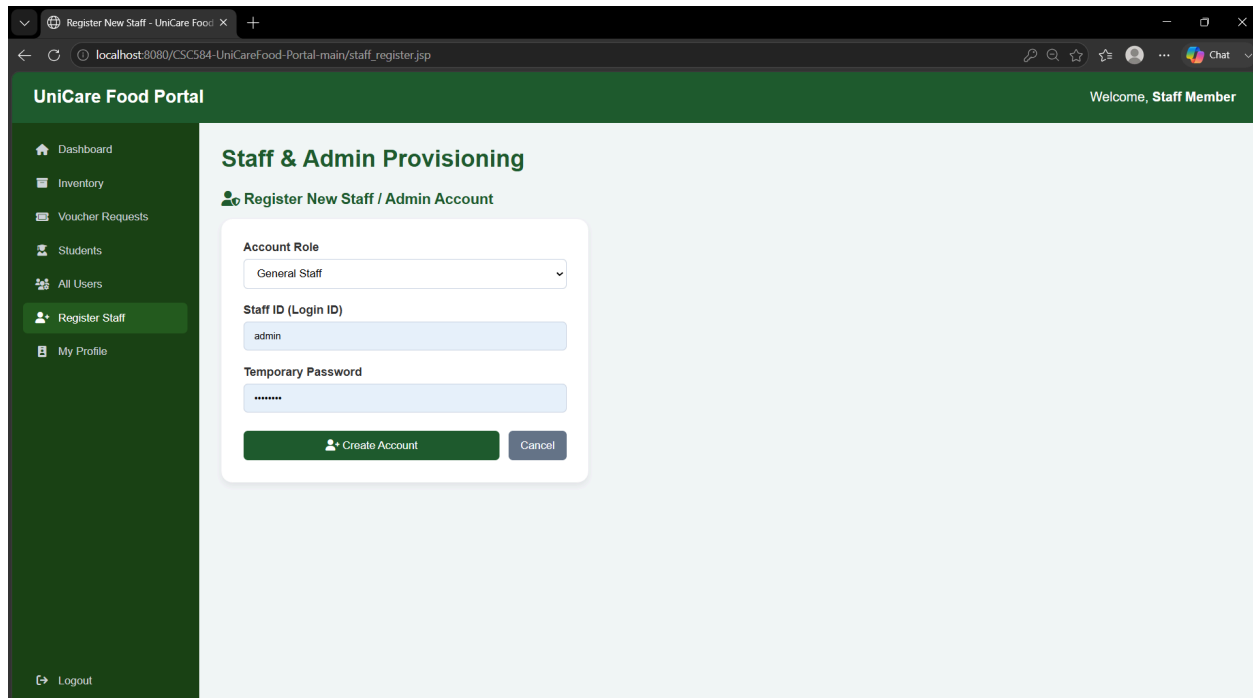
Account Role: You can select the specific operational authority level (such as General Staff or Admin) from the dropdown menu to define system access levels.

Staff ID (Login ID): You can input a unique alphanumeric identifier that the new team member will use to access their account.

Temporary Password: You can type a temporary password that the user will use for their initial portal entry.

Create Account Button: You can click the green Create Account button to save the record and officially authorize the new profile.

Cancel Button: You can click the grey Cancel button to discard your inputs and return to the main operational dashboard without saving.



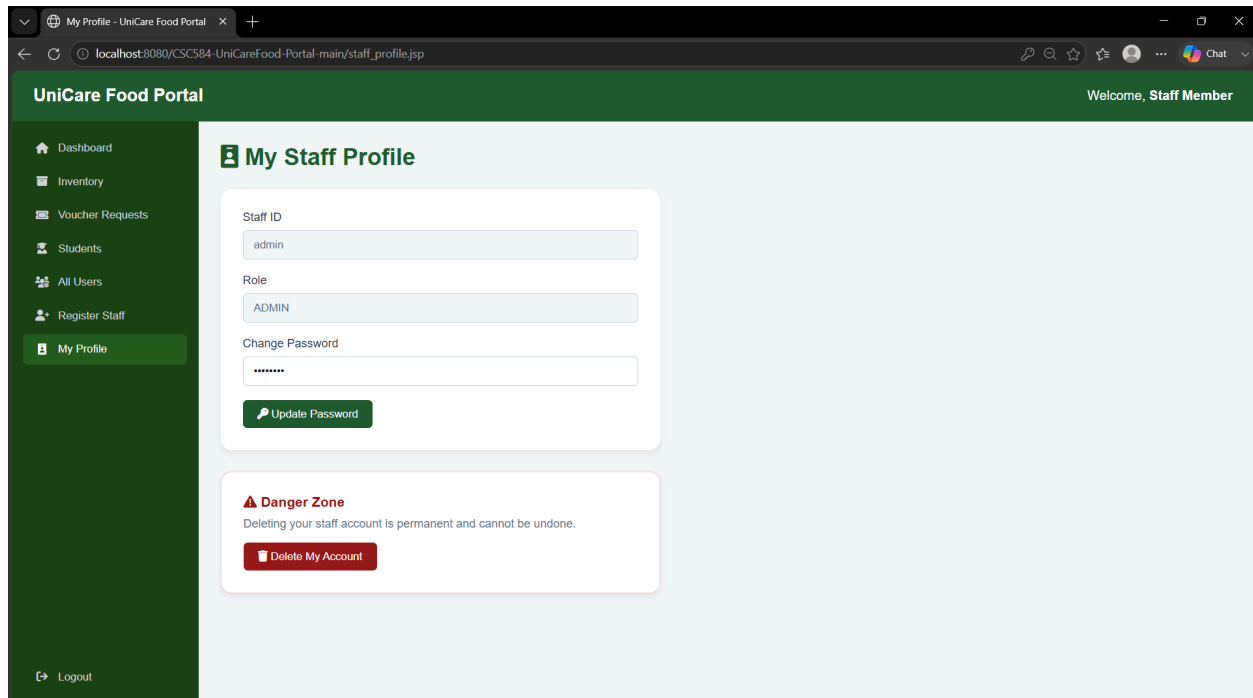
ix. Staff Profile Page

Profile Form Management

- **Staff ID:** You can see your locked administrative login username or identification key.
- **Role:** You can verify your active account authorization tier assigned within the portal.
- **Change Password:** You can type a new security phrase into this input field to update your current login credentials.
- **Update Password Button:** You can click the green Update Password button to save and apply your newly entered password.

Danger Zone Panel

- **Delete My Account:** You can click the red Delete My Account button to permanently remove your profile from the portal database.



Student Interface

x. Dashboard

Request a Food Voucher Panel

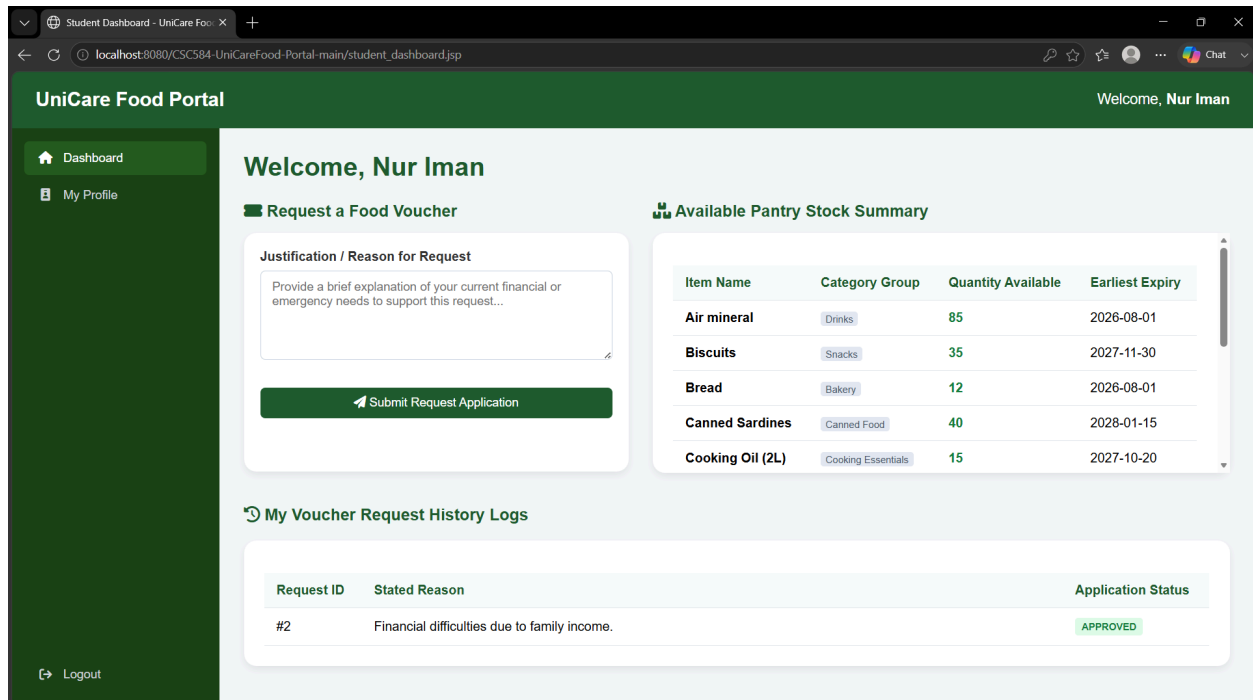
- **Justification / Reason for Request:** You can type a brief explanation of your current financial issues or emergency needs to support your application.
- **Submit Request Application Button:** You can click the green Submit Request Application button to send your voucher form to staff for administrative review.

Available Pantry Stock Summary

- **View Stock Lists:** You can browse active pantry supplies to monitor available resource items, category groupings, quantities remaining, and upcoming product expiration dates.

My Voucher Request History Logs

- **Track Request Profiles:** You can review all your previously submitted requests to monitor individual Request IDs, your stated justifications, and live status feedback badges (such as PENDING, REJECTED, and APPROVED).



xx. Student Profile Page

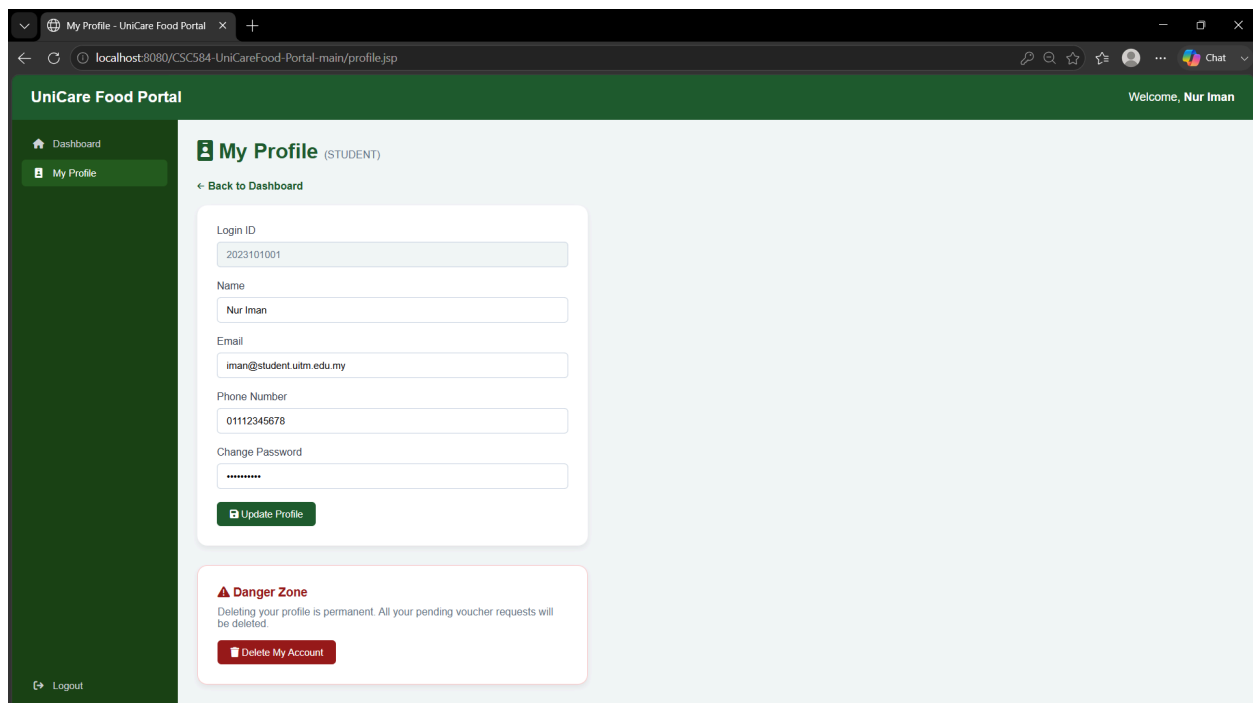
Profile Information Form

- **Back to Dashboard:** You can click this link to instantly return to your main student dashboard screen.
- **Login ID:** You can view your permanent student identification number used to log into the portal.
- **Name:** You can check or modify your full name registered inside the system.
- **Email:** You can review or edit your registered institutional email address.
- **Phone Number:** You can view or change your active mobile contact line.
- **Change Password:** You can type a new security phrase into this field if you wish to alter your account login password.

- **Update Profile Button:** You can click the green Update Profile button to save and apply any changes made to your profile fields or security password.

Danger Zone Panel

- **Delete My Account:** You can click the red Delete My Account button to permanently wipe your profile and automatically cancel all your pending voucher requests.

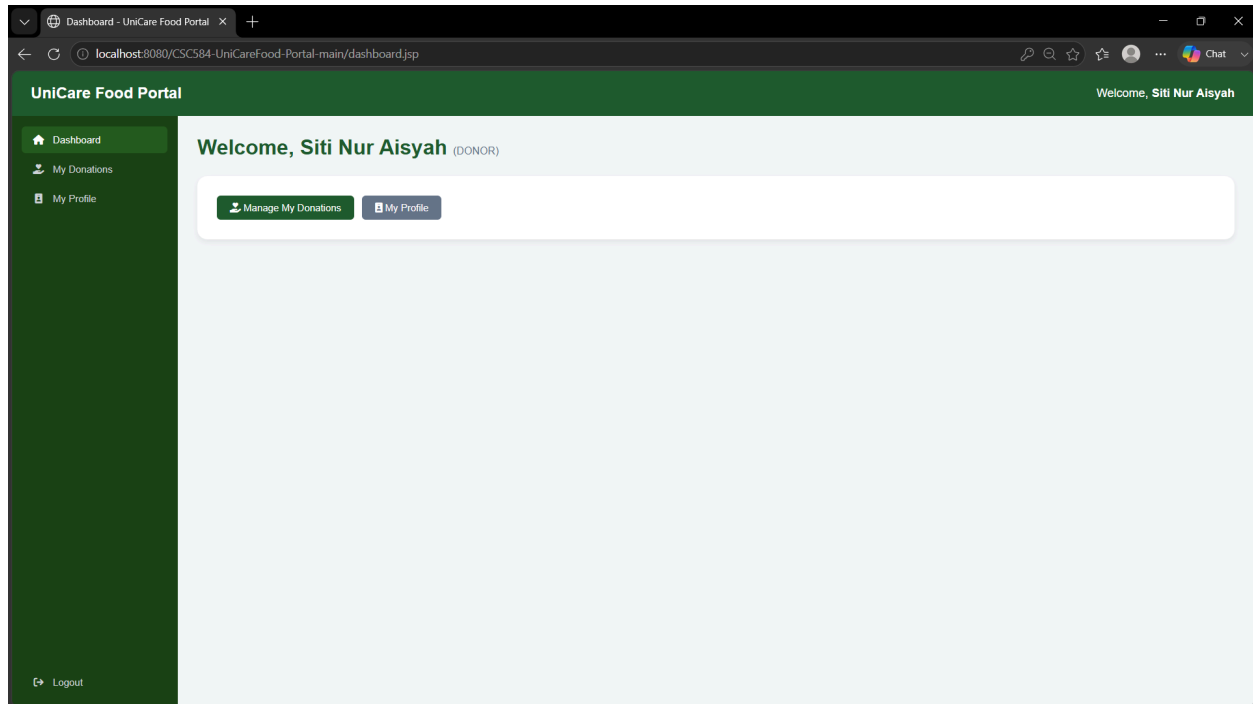


Donor Interface

xxx. Dashboard

Manage My Donations: You can click the green Manage My Donations button to view your active item history or log new food contributions into the portal pantry.

My Profile: You can click the grey My Profile button to access and edit your unique contact details, login keys, and security passwords.



xl. Donor Inventory Page

Donate New Item Form

- **Item Name:** You can type the descriptive name of the food item you want to contribute.
- **Category:** You can enter the specific food grouping that best matches your item.
- **Quantity:** You can input the total number of physical units you are donating.
- **Expiry Date:** You can select or type the exact expiration date for the product batch.
- **Add Donation Button:** You can click the green Add Donation button to register the new item into the portal tracking system.

My Donated Items List

- **Review Contributions:** You can view your active donation rows to check their assigned tracking ID, item details, category groups, stock quantities, and expiration dates.
- **Update Donation Details:** You can modify information fields directly in the row and click the green Update button to save changes to your donation entry.

- **Delete Record:** You can click the red Delete button to permanently cancel or remove an item record from your contribution list.

The screenshot shows the 'My Donations Workspace' interface. It includes a sidebar with 'Dashboard', 'My Donations', and 'My Profile'. The main area has a 'Donate New Item' form with fields for Item Name, Category, Quantity, and Expiry Date (dd/mm/yyyy), and an 'Add Donation' button. Below this is a 'My Donated Items' table with columns for ID, Item, Category, Quantity, Expiry Date, and Actions (Update, Delete).

ID	Item	Category	Quantity	Expiry Date	Actions
7	Cooking Oil (2L)	Cooking Essentials	15	20/10/2027	Update Delete
12	Bread	Bakery	12	01/08/2026	Update Delete

1. Donor Profile

Profile Information Form

- **Back to Dashboard:** You can click this link to instantly return to your main donor dashboard screen.
- **Login ID:** You can view your registered email address which acts as your permanent login identity key.
- **Name:** You can review or modify your full name or corporate identity recorded in the portal.
- **Phone Number:** You can view or change your active mobile or telephone contact line.
- **Change Password:** You can type a new security phrase into this field if you want to alter your portal account login password.

- **Update Profile Button:** You can click the green **Update Profile** button to save and apply any changes made to your data fields or account password.

Danger Zone Panel

- **Delete My Account:** You can click the red **Delete My Account** button to permanently wipe your profile and automatically make your previous donation listings anonymous.

The screenshot displays the 'My Profile' page of the UniCare Food Portal. The page features a dark green sidebar on the left with navigation links: 'Dashboard', 'My Donations', and 'My Profile'. The main content area has a light blue background. At the top of the main area, it says 'My Profile (DONOR)' with a 'Back to Dashboard' link. Below this is a white form containing the following fields: 'Login ID' (filled with 'siti@gmail.com'), 'Name' (filled with 'Siti Nur Aisyah'), 'Phone Number' (filled with '0138765432'), and 'Change Password' (masked with '*****'). A green 'Update Profile' button is positioned below the form. Underneath the form is a 'Danger Zone' section, which is highlighted with a red border and contains a warning icon, the text 'Deleting your profile is permanent. Your previous donations will become anonymous.', and a red 'Delete My Account' button. The top of the page has a green header with the text 'UniCare Food Portal' and 'Welcome, Siti Nur Aisyah'. The bottom left of the sidebar has a 'Logout' link.

Ix. Logout Page

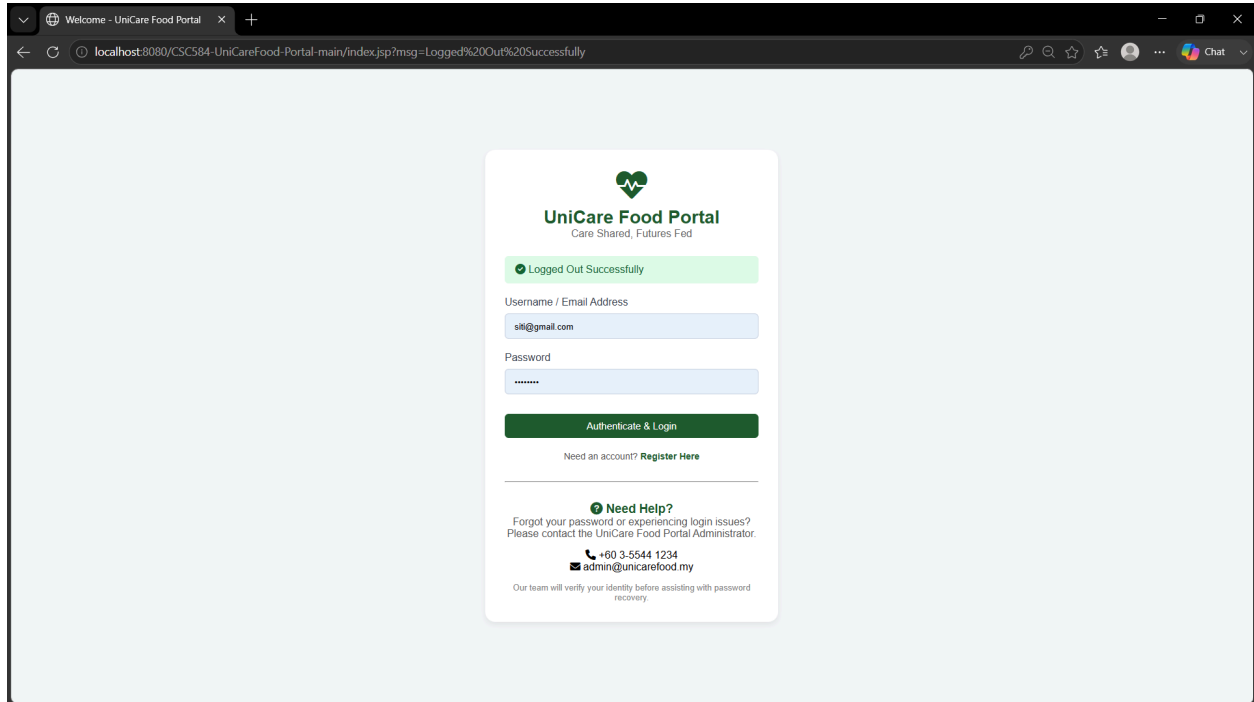
Login & Authentication Form

- **Logout Confirmation:** You can view a green confirmation banner verifying that you have logged out successfully from your previous session.
- **Username / Email Address:** You can type your registered portal username or donor email address into this field to start a new session.
- **Password:** You can type your secure account password into this input box.

- **Authenticate & Login Button:** You can click the green **Authenticate & Login** button to securely submit your portal credentials.
- **Register Here:** You can click the **Register Here** link if you do not have an account and want to sign up for portal access.

Need Help Support Panel

- **Administrator Contact Phone:** You can call the system support line at **+60 3-5544 1234** if you encounter system registration issues.
- **Support Email:** You can write an email to **admin@unicarefood.my** if you need assistance with password recovery or account locking policies.



The screenshot shows a web browser window with the address bar displaying "localhost:8080/CSC584-UniCareFood-Portal-main/index.jsp?msg=Logged%20Out%20Successfully". The page content is a login form for the "UniCare Food Portal" with the tagline "Care Shared, Futures Fed". At the top of the form, a green message box states "Logged Out Successfully". Below this, there are input fields for "Username / Email Address" (containing "siti@gmail.com") and "Password" (masked with dots). A green "Authenticate & Login" button is positioned below the password field. Underneath the button is a link that says "Need an account? Register Here". Further down, a "Need Help?" section provides contact information: "Forgot your password or experiencing login issues? Please contact the UniCare Food Portal Administrator." followed by a phone icon and "+60 3-5544 1234", and an email icon and "admin@unicarefood.my". At the very bottom, a small note reads: "Our team will verify your identity before assisting with password recovery."

6.0 Conclusion

The UniCare Food Portal was developed to optimize and simplify the administration of campus food assistance initiatives. Utilizing advanced Java Web Technology paired with a robust MVC architecture, the platform delivers a reliable, well-structured, and easily scalable user experience. For campus students, the portal accelerates the process of applying for food vouchers, checking application statuses, and reviewing live updates on pantry stock. Simultaneously, university staff benefit from a centralized interface that eliminates manual record-keeping errors, facilitates real-time inventory tracking, and clarifies the distribution workflow. The system also empowers philanthropic donors by offering a direct method to contribute items and update their donation logs seamlessly. By integrating these practical tools, the portal fosters a highly organized, transparent, and empathetic university environment, ensuring that vital aid is delivered precisely to those who need it most.